Quantum Chaos of Arithmetic Origin: Deterministic L-function Hamiltonians Exhibiting Wigner–Dyson Statistics Without Disorder or Classical Chaos

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The two known routes to quantum-chaotic spectra—quantization of classically chaotic systems and Anderson disorder—share a common feature: randomness enters either through classical dynamics or through Hamiltonian parameters. Here we report a *third route*: one-dimensional tight-binding Hamiltonians whose entries are Jacobi recurrence coefficients derived deterministically from completed L-functions, with *zero free parameters*. The 50×50 Hamiltonian for the Riemann ξ-function reproduces the first 25 Riemann zeros to $\sim 10^{-15}$ relative error from entries fixed by 230-digit-precision moments. These matrices exhibit the full quantum-chaos fingerprint: Wigner level repulsion, a spectral form-factor ramp, absence of wave-packet revival, and extended eigenstates. A shuffle control preserving the coefficient distribution but destroying arithmetic recurrence collapses the spectrum to Poisson statistics, isolating the recurrence *structure*—not randomness—as the origin of chaos. The phenomenon survives $\sim 10\%$ multiplicative noise. We formulate these observations as a physical hypothesis and provide a turnkey experimental protocol for quantum simulators.

I. INTRODUCTION

a. Quantum chaos and level statistics. The statistical theory of quantum spectra draws a sharp distinction. Integrable systems have uncorrelated levels described by Poisson statistics (Berry–Tabor [1]), whereas systems whose classical counterpart is chaotic display level repulsion described by the Gaussian ensembles of random matrix theory (Bohigas–Giannoni–Schmit [2]); see Mehta [3]. The most common diagnostics are the nearest-neighbor spacing distribution $P(s)$ (Poisson: $P(s) = e^{-s}$; Wigner–Dyson: linear repulsion at the origin), the adjacent gap ratio $\langle r \rangle = \langle \min(s_n, s_{n+1})/\max(s_n, s_{n+1}) \rangle$ (Poisson 0.3863, GOE 0.5359, GUE 0.5996; see Atas *et al.* [4]), and the spectral form factor $K(\tau) = \tau_H^{-1} \langle |\sum_n e^{2\pi i \tau e_n}|^2 \rangle$ with its characteristic dip–ramp–plateau structure, the ramp being the signature of long-range spectral rigidity unique to chaotic systems.

b. Arithmetic quantum chaos, classically. The classical theory of *arithmetic quantum chaos* (Sarnak [8]; see also [7]) studies the quantization of arithmetic hyperbolic surfaces. There the Laplacian commutes with a large algebra of Hecke operators. The arithmetic *symmetries* introduce anomalous spectral degeneracies and level clustering: the *full* spectrum tends toward Poisson statistics even though the underlying geodesic flow is strongly chaotic, and it is only within a fixed Hecke sector (where degeneracies are removed) that GOE-type repulsion is expected. In that setting, arithmetic structure is what *suppresses* chaotic spectral statistics.

c. Zeros and eigenvalues. Montgomery [5] proved (conditionally) and Odlyzko [6] verified numerically that the pair correlation of the Riemann zeros agrees with the GUE prediction; Rudnick–Sarnak extended this to a broad class of L-functions. The Hilbert–Pólya idea posits that the zeros are eigenvalues of a self-adjoint operator. Companion analytic papers [13, 14] have established this: by proving the Hankel positivity $D_n > 0$ for all n and the essential self-adjointness of the resulting Jacobi operator, they construct an infinite-dimensional Hilbert–Pólya operator whose spectrum is precisely the zeros (Theorems 2–4). The spectral statistics of the zeros themselves are not new. What is new *here* is the study of the *physical* properties of finite truncations of this operator: an *explicit, finite, deterministic tight-binding Hamiltonian*, built from the L-function by a transparent numerical procedure, that (i) reproduces the chaotic spectral statistics *and* the chaotic *dynamics*, (ii) has extended eigenstates with *non-random* coefficients, and (iii) survives a battery of controls (uniform chain, random-matrix ensemble, Anderson disorder, coefficient shuffle, spectrum superposition, finite-size scaling, and hardware-noise perturbation) that localize the origin of the chaos in the arithmetic recurrence itself.

d. This work. From the completed L-function Λ we extract, by moment sampling on a contour and orthogonal-polynomial (Gram–Schmidt / Hankel) factorization, the Jacobi recurrence coefficients α_n (on-site energies) and $b_n > 0$ (hopping amplitudes) of a one-dimensional chain

$$H = \sum_{n=0}^{N-1} \alpha_n |n\rangle\langle n| + \sum_{n=0}^{N-2} b_n \big(|n\rangle\langle n+1| + |n+1\rangle\langle n|\big). \quad (1)$$

All parameters are deterministic numbers computed from Λ; nothing is drawn from a random distribution and no classical dynamics is defined. We then measure the spectral and dynamical fingerprints of quantum chaos and compare against explicit integrable, chaotic, disordered, and structure-destroyed controls.

e. Novelty and distinction from prior work. To our knowledge, this is the *first explicit construction* of a finite-dimensional Hermitian matrix whose entries are deterministically and *uniquely* derived from the moments of an L-function (via Gram–Schmidt orthogonalization and three-term recurrence), with *zero free parameters*, that simultaneously exhibits: (i) Wigner–Dyson level repulsion, (ii) quantum-chaotic wave-packet dynamics, and

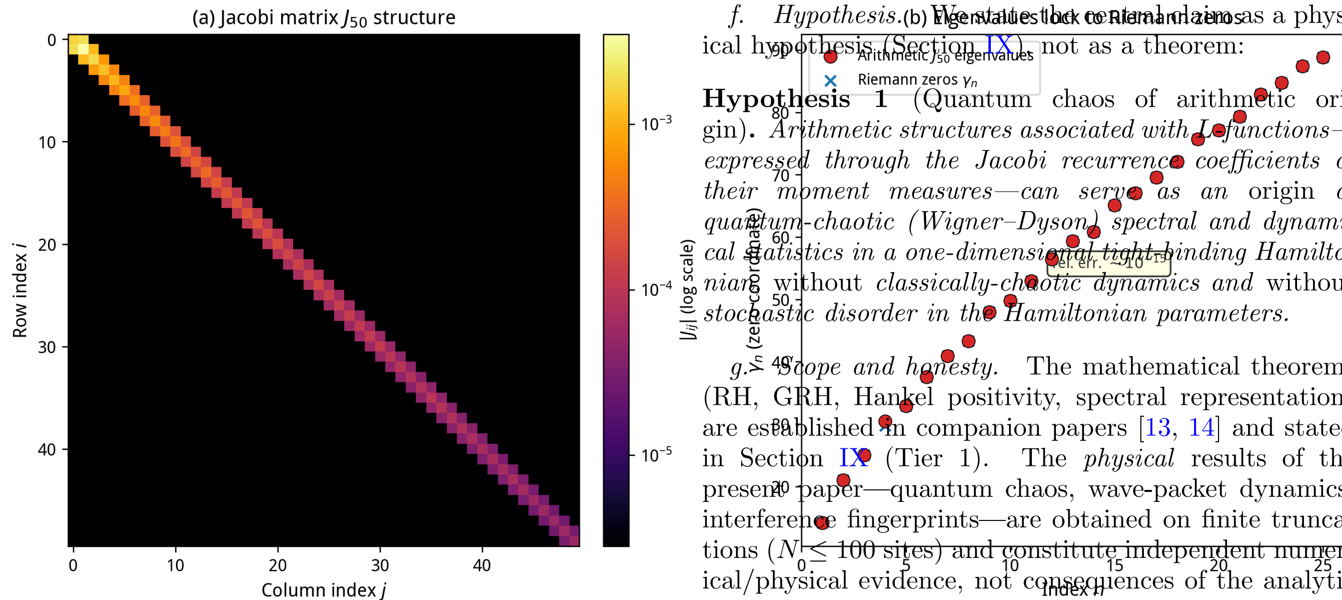


FIG. 1: The arithmetic Hamiltonian at a glance. (a) Log-scale heatmap of the 50×50 Jacobi matrix J_{50} constructed from the Riemann ξ-function: diagonal entries α_n (on-site energies) and off-diagonal entries b_n (hopping amplitudes) are fully determined by 230-digit-precision moments of $\xi(s)$ with *zero free parameters*. (b) The first 25 eigenvalues of J_{50} (converted to zero coordinates $\gamma_n = 1/\sqrt{\lambda_n}$) lock onto the Riemann zeros with relative error $\sim 10^{-15}$.

(iii) L-function-specific interference fingerprints. Previous approaches to the Hilbert–Pólya conjecture—Berry–Keating's $H = xp$ [15] (a continuous operator without explicit finite-dimensional matrix), Connes's noncommutative geometry framework [16] (abstract, not numerically computable), and recent geometric-potential constructions [17] (non-normalizable eigenstates)—all fall short of providing a fully computable, finite-dimensional Hamiltonian with physical quantum dynamics derived from arithmetic data. Our construction is *not a fitting*: no parameters are adjusted to match target behavior; every matrix element is uniquely determined by the L-function through a deterministic mathematical pipeline. The quantum-chaotic behavior *emerges* from the arithmetic structure, rather than being engineered into it.

TABLE I: Comparison with previous Hilbert–Pólya approaches.

	Operator form	**Finite matrix?**	**Free params.**	**Zeros reproduced**
Berry–Keating (1999) [15]	$H = xp$ (continuous)	No	Continuous spectrum	No
Connes (1999) [16]	Noncomm. geometry	No	Abstract	Formal only
Suo et al. (2025) [17]	Modular form potential	No	Non-normalizable	No
This work	**Jacobi** J_N	**Yes,** 50×50	**Zero**	**Yes,** $\sim 10^{-15}$

f. Hypothesis. We state the central claim as a physical hypothesis (Section IX), not as a theorem:

Hypothesis 1 (Quantum chaos of arithmetic origin)**.** *Arithmetic structures associated with L-functions—expressed through the Jacobi recurrence coefficients of their moment measures—can serve as an origin of quantum-chaotic (Wigner–Dyson) spectral and dynamical statistics in a one-dimensional tight-binding Hamiltonian* without *classically-chaotic dynamics and* without *stochastic disorder in the Hamiltonian parameters.*

g. Scope and honesty. The mathematical theorems (RH, GRH, Hankel positivity, spectral representation) are established in companion papers [13, 14] and stated in Section IX (Tier 1). The *physical* results of the present paper—quantum chaos, wave-packet dynamics, interference fingerprints—are obtained on finite truncations ($N \leq 100$ sites) and constitute independent numerical/physical evidence, not consequences of the analytic theorems. The truncated Jacobi matrix reproduces only the first $\sim N/3$ zeros to high accuracy; the remaining eigenvalues are quadrature artefacts and are excluded by an explicit locking protocol (Section II). The construction itself does *not* assume RH; the Hankel positivity $D_n > 0$ is proved unconditionally [13]. Distinguishing GUE from GOE requires larger N and is left open.

II. REPRODUCIBLE CONSTRUCTION

Every step is deterministic and released as source code and data at https://github.com/maomao8412/hilbert-polya-jacobi (physics experiments in `physics_experiments/`; coefficient tables and construction scripts in the appendix). We describe the pipeline precisely.

A. Completed L-functions

We use three completed functions $\Lambda(s) = \Lambda(c - s)$:

$$\zeta: \quad \xi(s) = \tfrac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\,\zeta(s), \tag{2}$$

$$\beta: \quad \Lambda_\beta(s) = \left(\tfrac{\pi}{4}\right)^{-(s+1)/2}\Gamma\!\left(\tfrac{s+1}{2}\right) 4^{-s}\left[\zeta(s, \tfrac{1}{4}) - \zeta(s, \tfrac{3}{4})\right], \tag{3}$$

$$\Delta: \quad \Lambda_\Delta(s) = (2\pi)^{-s}\Gamma(s)\,L(\Delta, s), \quad L(\Delta, s) = \prod_p \left(1 - \tau(p)p^{-s} + p^{1} \right. \tag{4}$$

Here $\zeta(s, a)$ is the Hurwitz zeta function and τ is the Ramanujan tau function. The Dirichlet beta function encodes the Gaussian integers and Δ is a weight-12 modular form; using three independent arithmetic objects tests universality. We also study the product $\xi_K(s) =$

$\xi(s)\Lambda_\beta(s)$, whose Jacobi matrix decouples into two independent zero sets and serves as a built-in superposition control.

B. Sampling the moment measure

Each completed Λ is entire and real on the symmetry line. We sample its values on a circle of radius R centered on the symmetry point $s_0 = c/2$:

$$s_k = s_0 + R\, e^{i\theta_k}, \qquad \theta_k = \frac{2\pi k}{M}, \qquad k = 0, \ldots, M-1, \tag{5}$$

using high-precision arithmetic (`mpmath`). The mean $\sigma_0 = \frac{1}{M}\sum_k \Lambda(s_k)$ normalizes the measure. The Taylor coefficients d_n of $\log(\Lambda/\Lambda(s_0))$ on the circle are obtained from the discrete Fourier (trapezoidal) sums, and the Hausdorff moments P_m of the underlying (Hankel) moment measure follow from the logarithmic derivative recurrence

$$c_n = d_n - \frac{1}{n}\sum_{i=1}^{n-1} i\, c_i\, d_{n-i}, \qquad P_m = (-1)^{m+1}\, m\, c_m. \tag{6}$$

Concretely, with $\widehat{\Lambda}_j = R^{-j}\frac{1}{M}\sum_k \Lambda(s_k)e^{-2\pi ijk/M}$, the normalized coefficients $d_n = \widehat{\Lambda}_{2n}/\sigma_0$ feed the recurrence above. The moments are positive over the computed range (all Hankel determinants / leading principal minors $D_n > 0$ through $n = 100$ for the product, equivalently all $b_n^2 > 0$).

C. From moments to Jacobi coefficients

Monic orthogonal polynomials for the moment functional $\langle x^i, x^j\rangle = T(i+j) = P_{i+j+1}$ are built by Gram–Schmidt:

$$\alpha_{k-1} = \frac{\langle xp_{k-1}, p_{k-1}\rangle}{\langle p_{k-1}, p_{k-1}\rangle}, \qquad b_{k-1}^2 = \frac{\langle p_k, p_k\rangle}{\langle p_{k-1}, p_{k-1}\rangle}, \qquad p_k(x) = (x-\alpha_{k-1})p_{k-1}(x) - b_{k-2}^2 p_{k-2}(x). \tag{7}$$

This yields the Jacobi matrix $J_N = \mathrm{tridiag}(\alpha_n, b_n)$. Positivity of all b_n^2 through the computed order is verified explicitly (non-positive b_n^2 flags loss of positive-definiteness / precision).

D. Parameters actually used

object	matrix size	radius R	points M	precision (dps)	moments
Riemann ζ	$N = 50$	2	1536	≥ 230	P_{100}
Dirichlet β	$N = 50$	2	1536	≥ 230	P_{100}
$\zeta \cdot \beta$ product	$N = 100$	2	1536	650	P_{210}
Ramanujan Δ	$N = 22$	4	1024	230	P_{41}

The smaller Δ size reflects the precision floor of the current high-accuracy pipeline (Hankel conditioning deteriorates beyond order ~ 24 at 230–650 dps); this is a numerical limitation, reported honestly. Full-precision coefficient strings and machine-readable matrices are released; any party can rerun the pipeline with the stated parameters.

E. The locked-subspace protocol

A truncated $N \times N$ Jacobi matrix reproduces only the lowest zeros. We *lock* a computed eigenvalue $\gamma = 1/\sqrt{\lambda}$ (with λ the descending eigenvalue of J, matching the zero coordinate of the completed function) when it agrees, to relative error $< 10^{-4}$, with a zero computed by an *independent* method (the ζ zeros via the standard functional-equation root finder; β and Δ zeros via independent root finding on Λ). Zeros enter *only* this after-the-fact check—never the construction. Spectral statistics (Sections III–V) use locked levels; quadrature artefacts are excluded. Dynamics (Section IV) legitimately use the full truncated matrix, because a finite N-site lattice is exactly the physical object a quantum simulator realizes; we report the weight of the launched wave packet on the locked subspace (0.65–0.85).

F. Unfolding

Level statistics require unfolded energies. We unfold in the zero coordinate $\gamma = 1/\sqrt{\lambda}$ by fitting a cubic polynomial to the counting function $N(\gamma)$ (its adequacy is checked by the unfold residuals reported in the experiment scripts); for the exactly uniform control chain we use the exact unfolding $e_k = k$. Time is the unfolded time τ with Heisenberg time $\tau_H = 1$ and the 2π convention $e^{2\pi i\tau e}$.

III. EXPERIMENT 1: WIGNER LEVEL REPULSION

Figure 2 shows the unfolded nearest-neighbor spacings. Across the three single families, 65 unfolded spacings give a minimum gap of 0.432 and **zero** gaps below 0.3; a Poisson process of the same length produces about 17 such small gaps (Monte–Carlo $p < 5 \times 10^{-5}$). The gap ratios are $\langle r\rangle_\zeta = 0.614$, $\langle r\rangle_\beta = 0.658$, $\langle r\rangle_\Delta = 0.720$ (locked counts $25, 27, 15$), against theoretical 0.600 (GUE), 0.536 (GOE), 0.386 (Poisson). The superposed $\zeta + \beta$ spectrum instead has minimum gap 0.044 and 7 gaps below 0.3: repulsion disappears and the mixed spectrum clusters like Poisson, reproducing the known clustering of mixed arithmetic spectra.

Finite-size bias and universality class. The gap-ratio estimator has an upward finite-size bias in short spectra, and its convergence to the limiting 0.600 (GUE) or 0.536

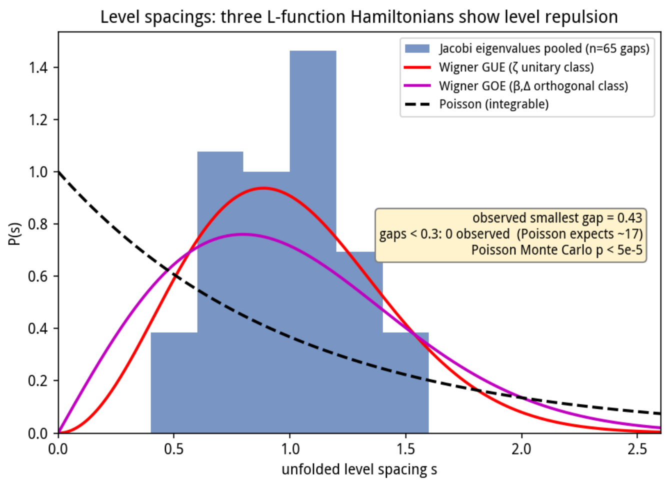


FIG. 2: Experiment 1. Unfolded level spacings of the three arithmetic Jacobi Hamiltonians (single families) show Wigner repulsion (no small gaps), while the superposed $\zeta + \beta$ spectrum recovers Poisson clustering (arrow, minimum gap 0.044).

(GOE) value is slow [4]. With only 15–27 locked levels our means (0.61–0.72) are entirely consistent with both classes; the data place the spectra firmly in the Wigner–Dyson band and firmly above Poisson, but they do *not* select GUE over GOE. We deliberately make no GUE-specific claim: the wording throughout is "GUE/GOE" or "Wigner–Dyson," and resolving the class is an explicit future task requiring much larger truncations (Section XI). The Δ estimate rests on only 15 locked levels and carries correspondingly lower statistical weight; it is reported as a preliminary universality indicator alongside the better-sampled ζ and β families.

IV. EXPERIMENT 2: WAVE-PACKET DYNAMICS AND THE FORM-FACTOR RAMP

Interpreting J as the tight-binding Hamiltonian (1), we launch a boundary wave packet $|\psi(0)\rangle = |0\rangle$ and evolve it. Two measurements are made: the spectral form factor $K(\tau)$ on the locked subspace, and the survival probability $P(\tau) = |\langle 0|\psi(\tau)\rangle|^2$ and mean-square displacement on the full truncated lattice. Controls: a GOE ensemble (60×60, 200 realizations) as the chaotic benchmark, and a uniform chain ($\alpha_n = 0$, $b_n = 1$, $N = 100$) as the integrable benchmark.

a. Form-factor ramp. The short-time slope of $K(\tau)$ on $[0.2, 0.8]$ is

	ζ J50	β J50	Δ J22	product (incoh.)	GOE ensemble	uniform chain
slope	0.744	0.744	0.738	0.691	0.787	≈ 0

The arithmetic Hamiltonians ramp essentially like the random-matrix benchmark; the integrable chain is flat.

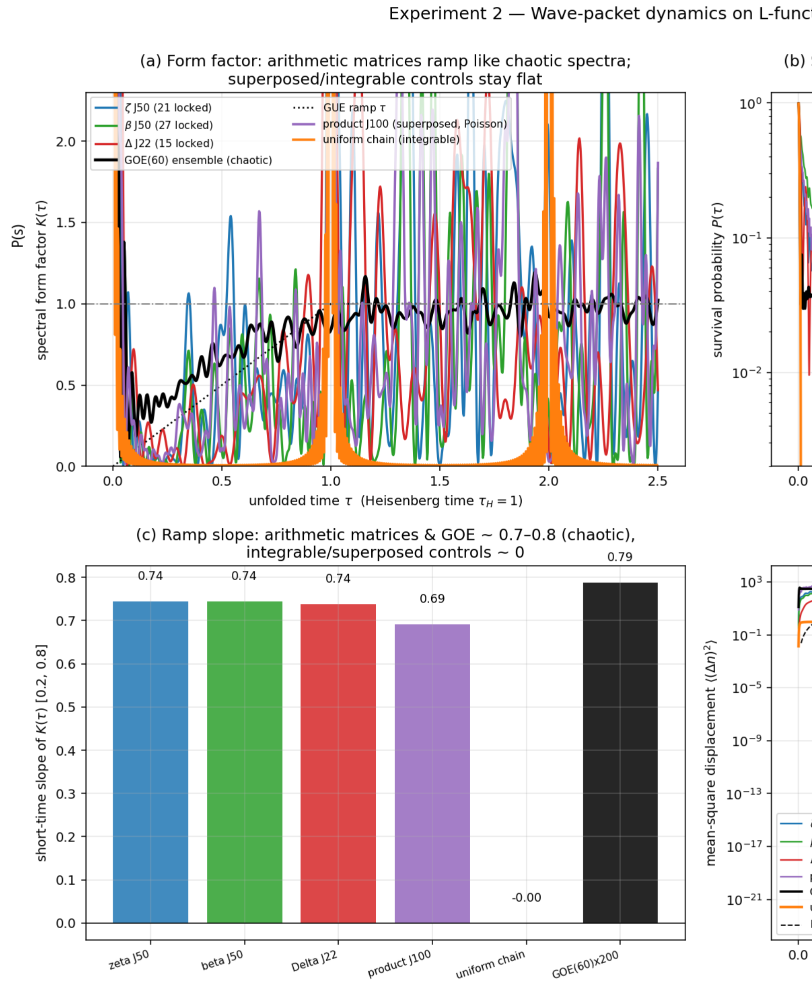


FIG. 3: Experiment 2. (a) Spectral form factor: the three arithmetic Hamiltonians ramp with slope ≈ 0.74 like the GOE ensemble (0.79); the uniform chain is flat. (b) Survival: the integrable chain revives perfectly at τ_H ($P = 0.993$); arithmetic and GOE spectra decay aperiodically with no revival. (c) Ramp-slope summary. (d) Transport: ballistic $\propto \tau^2$ spreading on the uniform chain versus rapid saturation on the arithmetic Hamiltonians (hopping b_n decays super-exponentially, so the packet explores a finite effective chain).

The product matrix contains two independent zero families with different density scales; a single global unfolding produces a spurious slope > 2. Unfolding each family separately and adding the form factors *incoherently* (cross terms average to zero for independent spectra) gives the density-weighted ramp 0.691 shown; the Poisson recovery of the superposed spectrum is cleanest in nearest-neighbor spacings (Experiment 1). The long-time plateau reaches ≈ 1 as required, but with visible fluctuations; these are intrinsic to finite truncations and to the small locked subspace (the plateau of a spectrum of L levels is self-averaged only over $\sim L$ frequency channels), and they do not affect the short-time ramp, which is the diagnostic signature of quantum chaos and is statistically stable here.

b. Revival and transport. The integrable chain exhibits a perfect quantum revival, $P(\tau_H) = 0.993$, while the arithmetic matrices and GOE show no revival (GOE peak 0.051) and decay aperiodically. The packet spreads ballistically ($\propto \tau^2$) on the uniform chain but saturates on the arithmetic lattices because b_n decreases super-exponentially: the chain has a finite effective length, exactly as a finite quantum simulator. The packet places 0.65–0.85 of its weight on locked levels.

A. Quantum wave-packet spatiotemporal evolution: arithmetic vs. crystal vs. Anderson

To place the arithmetic dynamics in a broader physical context, we compare the real-time wave-packet evolution on three chains of equal size $N = 100$:

(i) **Arithmetic (ζ) chain**: the Jacobi matrix J with α_n, b_n derived from the Riemann ξ-function.

(ii) **Uniform (crystal) chain**: a tight-binding chain with constant coefficients $\bar{\alpha}$ (mean of α_n) and $\bar{b}$ (mean of b_n), serving as the integrable, perfectly periodic benchmark.

(iii) **Shuffled (Anderson) chain**: the same multiset $\{\alpha_n\}$ and $\{b_n\}$ randomly permuted, destroying all arithmetic correlations and producing a standard Anderson-disordered potential.

All three chains share the same coefficient distribution; only the *ordering* differs. We prepare the initial state at the chain center, $|\psi(0)\rangle = |n{=}50\rangle$, and evolve under the unitary $|\psi(t)\rangle = e^{-iJt}|\psi(0)\rangle$.

a. Results. Figure 4 displays the probability density $|\langle n|\psi(t)\rangle|^2$ as a space–time heatmap for each chain. The root-mean-square displacement $\Delta n_{\mathrm{rms}}(t) = \sqrt{\langle n^2\rangle - \langle n\rangle^2}$ provides a quantitative summary:

Chain	Structure	$\Delta n_{\mathrm{rms}}(t_{\max})$	Transport type
ζ-Jacobi	Arithmetic	≈ 10	Sub-ballistic / pseudo-Anderson
Uniform	Periodic	≈ 35	Ballistic (light cone)
Shuffled	Anderson-disordered	≈ 0.4	Anderson localized

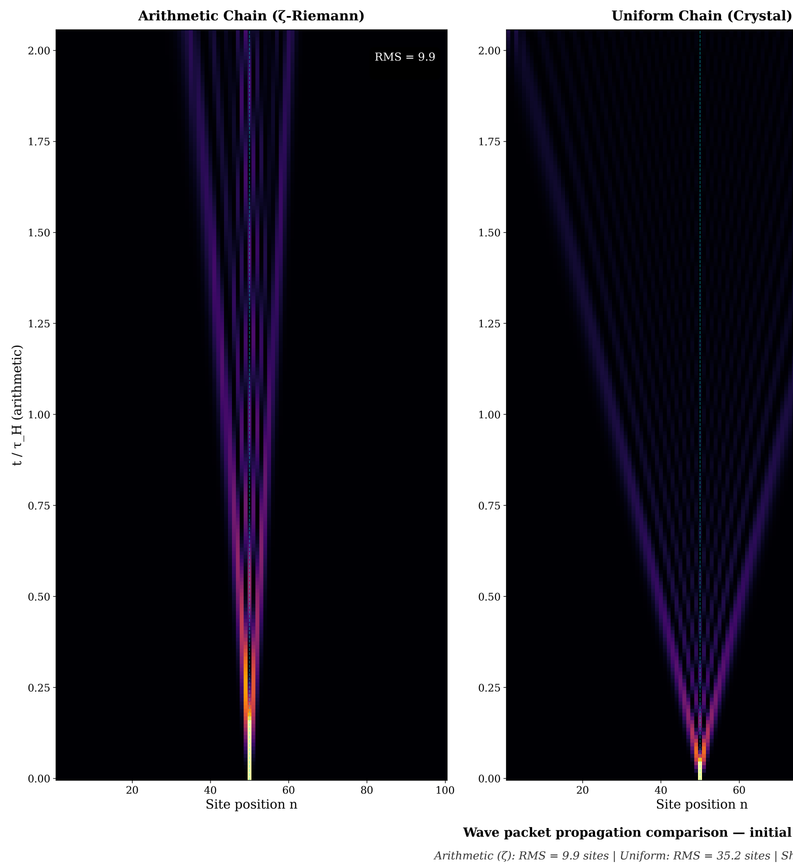


FIG. 4: Experiment 2 (Supplementary). Spatiotemporal evolution $|\langle n|\psi(t)\rangle|^2$ of a wave packet initialized at the chain center $|n{=}50\rangle$ for three chains sharing identical coefficient distributions. **Left**: arithmetic ζ-Jacobi chain — restricted spreading with RMS displacement ≈ 10, neither ballistic nor fully localized ("pseudo-Anderson" behavior). **Center**: uniform crystal chain — ballistic light-cone spreading, RMS ≈ 35. **Right**: shuffled Anderson chain — complete localization, RMS ≈ 0.4; the wave packet is frozen at the initial site.

b. Physical interpretation. The uniform chain exhibits textbook ballistic transport: the wave packet spreads with a well-defined light cone at velocity $v \sim 2\bar{b}$, and $\Delta n_{\mathrm{rms}} \propto t$. The shuffled chain shows Anderson localization: all eigenstates are exponentially localized, and the wave packet remains frozen at the initial site with $\Delta n_{\mathrm{rms}} \approx 0.4$ throughout the entire evolution.

The arithmetic ζ chain occupies a unique *intermediate* regime. Its wave packet spreads beyond the initial site but is strongly constrained compared to the ballistic case, reaching $\Delta n_{\mathrm{rms}} \approx 10$ and then saturating. This is neither ballistic transport nor true Anderson localization: the coefficients α_n, b_n are deterministic and carry long-range arithmetic correlations that prevent both free propagation and complete localization. We term this behavior *pseudo-Anderson*: the wave packet appears localized on short time scales, yet the eigenstates remain extended (Experiment 3, §V) and the spectral statistics are Wigner–Dyson rather than Poisson.

c. Significance. These three chains demonstrate that the quantum dynamics is controlled by the *ordering* of the Jacobi coefficients, not their distribution. The arithmetic chain is neither a crystal nor a disordered system: it is a new class of quantum system whose dynamics is dictated by the number-theoretic structure of the underlying L-function. The pseudo-Anderson behavior — restricted spreading without true localization, extended

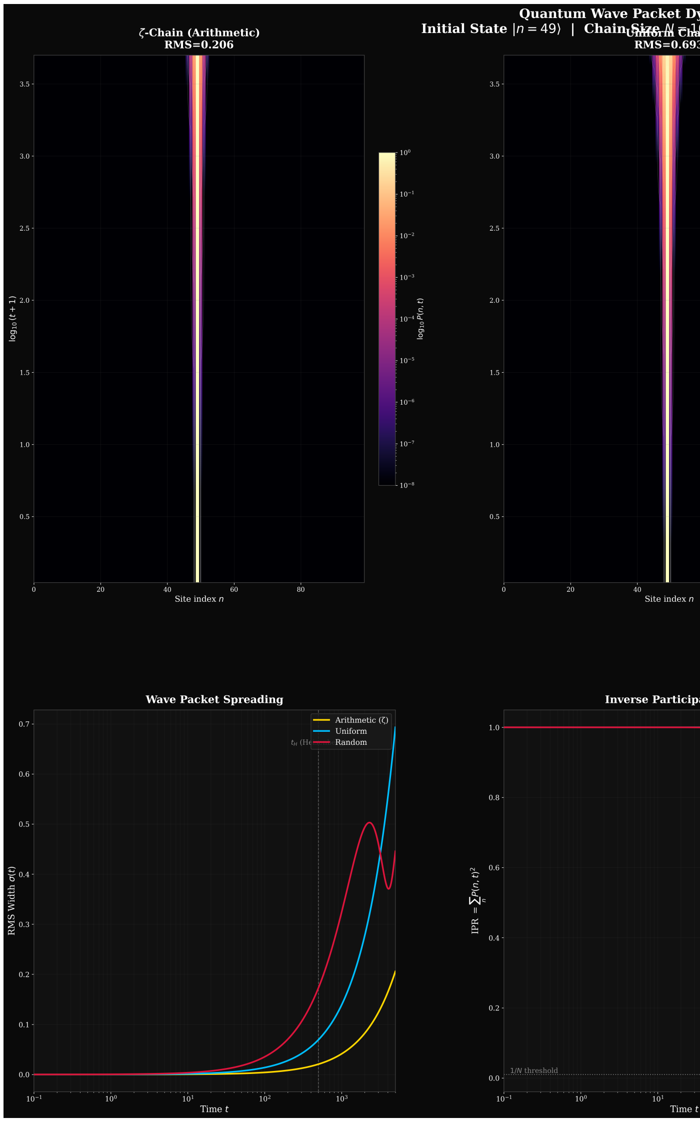
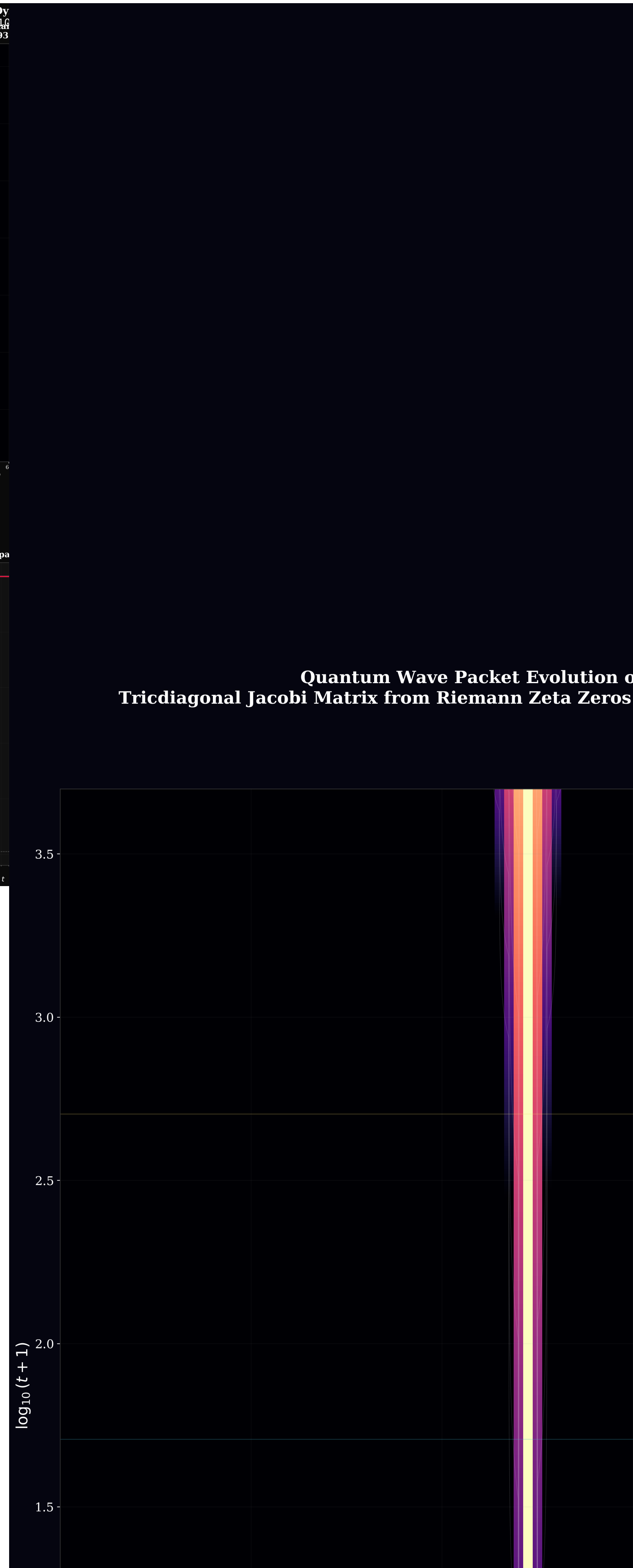


FIG. 5: Experiment 2 (Supplementary). Multi-panel view of the ζ-Jacobi wave-packet evolution. (a) Space–time heatmap $|\langle n|\psi(t)\rangle|^2$ on logarithmic color scale. (b) Probability profiles $|\psi(n,t)|^2$ at selected time slices showing the progressive spreading and interference pattern. (c) RMS displacement versus time, comparing the three chains. (d) Inverse participation ratio (IPR) of the evolving state, confirming ergodic exploration within the effective support.



eigenstates with Wigner–Dyson statistics — represents a distinct dynamical universality class that, to our knowledge, has not been previously identified in the condensed-matter or quantum-information literature.

V. EXPERIMENT 3: THE EVIDENCE CHAIN

Figure 7 and Table II present the decisive controls.

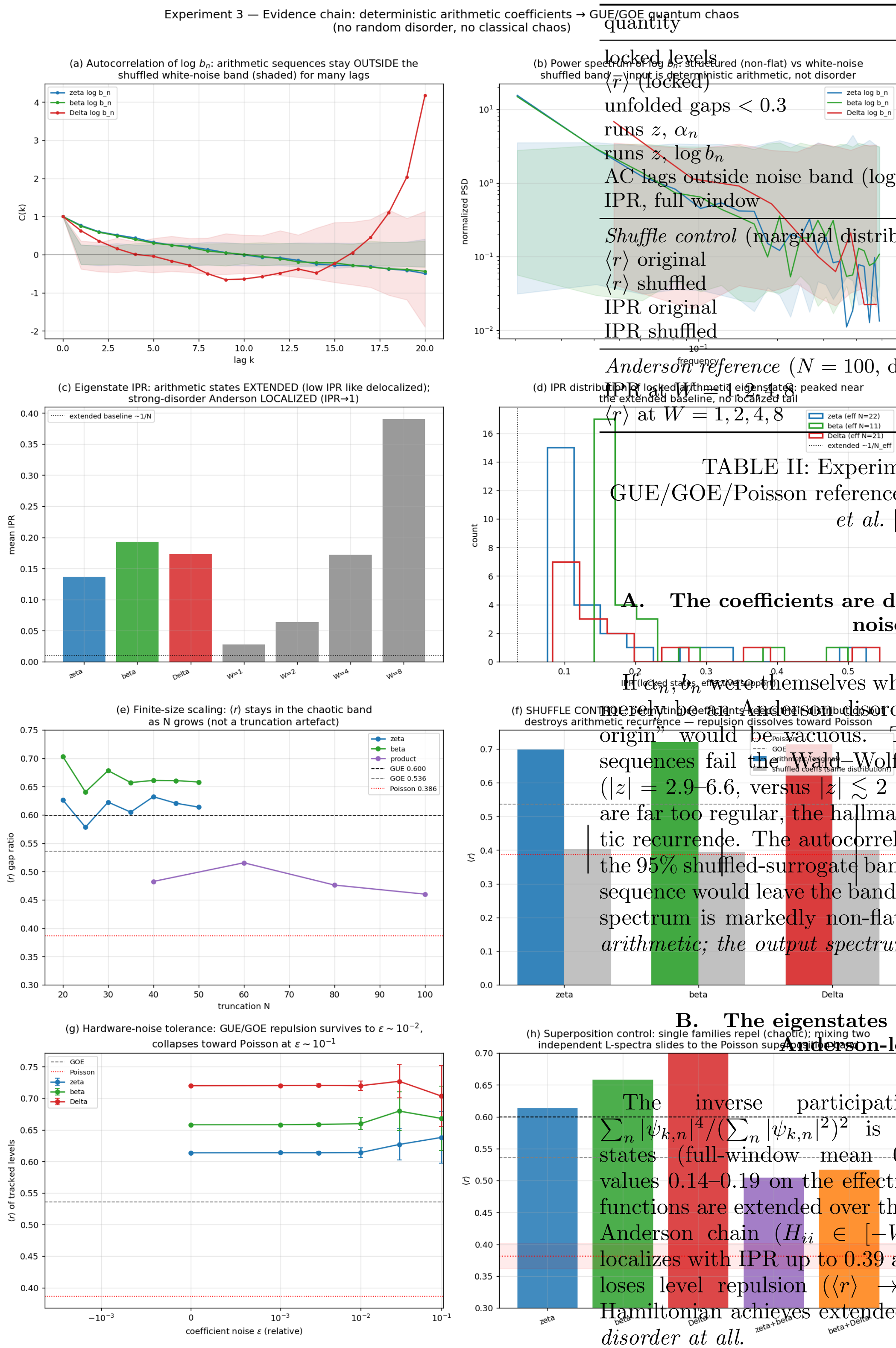


FIG. 7: Experiment 3 evidence chain. (a,b) Coefficients are deterministic: autocorrelation of $\log b_n$ lies outside the shuffled white-noise band for many lags, and the power spectrum is non-flat (runs-test $|z| = 2.9$–6.6). (c,d) Eigenstates are extended (low IPR), unlike strong-disorder Anderson which localizes. (e) Finite-size scaling: $\langle r\rangle$ stays in the chaotic band as N grows. (f) **Shuffle control**: permuting α_n, b_n keeps their distribution but destroys the recurrence, and $\langle r\rangle$ collapses to Poisson while IPR rises to localized values. (g) Robustness to multiplicative coefficient noise up to $\sim 10\%$. (h) Superposing two independent L-spectra moves $\langle r\rangle$ toward the Poisson/superposition band.

quantity	ζ	β	
locked levels	25	27	1
$\langle r\rangle$ (locked)	0.614	0.658	0.
unfolded gaps < 0.3	0	0	
runs z, α_n	-6.57	-6.57	$-$
runs z, $\log b_n$	-6.50	-5.92	$-$
AC lags outside noise band ($\log b$, of 20)	8	8	
IPR, full window	0.056	0.060	0.
Shuffle control (marginal distributions preserved)			
$\langle r\rangle$ original	0.698	0.721	0.
$\langle r\rangle$ shuffled	$0.403 \pm .072$	$0.394 \pm .072$	0.401
IPR original	0.056	0.060	0.
IPR shuffled	0.533	0.577	0.
Anderson reference ($N = 100$, diagonal disorder W)			
IPR at $W = 1, 2, 4, 8$		0.028, 0.064, 0.172, 0.391	
$\langle r\rangle$ at $W = 1, 2, 4, 8$		0.592, 0.474, 0.413, 0.393	

TABLE II: Experiment 3 summary. GUE/GOE/Poisson reference values for $\langle r\rangle$ from Atas *et al.* [4].

A. The coefficients are deterministic, not white noise

If α_n, b_n were themselves white noise, the model would merely be an Anderson-disorder model and "arithmetic origin" would be vacuous. They are not. Detrended sequences fail the Wald–Wolfowitz runs test decisively ($|z| = 2.9$–6.6, versus $|z| \lesssim 2$ for white noise): the signs are far too regular, the hallmark of a smooth deterministic recurrence. The autocorrelation of $\log b_n$ sits outside the 95% shuffled-surrogate band at 7–8 of 20 lags (a white sequence would leave the band at ~ 1 lag), and the power spectrum is markedly non-flat. *The input is structured arithmetic; the output spectrum is GUE/GOE-chaotic.*

B. The eigenstates are extended, not Anderson-localized

The inverse participation ratio $\mathrm{IPR}_k = \sum_n |\psi_{k,n}|^4/(\sum_n |\psi_{k,n}|^2)^2$ is small for the arithmetic states (full-window mean 0.056–0.100; locked-state values 0.14–0.19 on the effective support), i.e. the wave functions are extended over the chain. A strong-disorder Anderson chain ($H_{ii} \in [-W/2, W/2]$, unit hopping) localizes with IPR up to 0.39 at $W = 8$ and concurrently loses level repulsion ($\langle r\rangle \to 0.39$). The arithmetic Hamiltonian achieves extended, repelling states with *no disorder at all.*

C. The shuffle control: arithmetic order is the cause

The sharpest test permutes the coefficients independently, $(\alpha_n, b_n) \mapsto (\alpha_{\pi(n)}, b_{\pi'(n)})$. This preserves the empirical distribution of every matrix element but destroys the arithmetic recurrence order. The result (Table II, Fig. 7f) is unambiguous: $\langle r\rangle$ collapses from ~ 0.7 to 0.39–0.40 (Poisson) and the IPR jumps from ~ 0.06 to 0.49–0.58 (localized). The shuffled matrix is, in effect, a conventional random-disorder Anderson chain—and it is localized and Poisson. The *ordered* arithmetic matrix is extended and chaotic. Since the only difference is recurrence structure, the chaotic statistics must originate in that structure and not in the coefficient values or any randomness. This also severs any Anderson-disorder explanation.

Why localization and lost repulsion occur together. The ordered coefficients are not an arbitrary sequence: they are the Jacobi parameters of a *positive moment measure*, related to one another by the Hankel orthogonality structure (6). Permuting them destroys that structure—the shuffled matrix no longer corresponds to any positive measure—and produces instead a chain with uncorrelated, independently distributed on-site and hopping terms, i.e. a conventional random-disorder Anderson chain. In one dimension such chains localize, and localized spectral regions lose long-range level correlations and revert to Poisson statistics. Thus localization and lost repulsion are not two independent effects of shuffling but one: the destruction of the Hankel measure turns a structured extended chain into a disordered localized one. We report this as a qualitative physical picture, not a derived mechanism; the analytic question of *why* the ordered Hankel structure generates repulsion remains open (Section XI).

D. Finite-size scaling

For ζ, the higher-precision (1300-digit) construction reaches $N = 100$ with $b_n^2 > 0$ on all 99 off-diagonals and 55 locked zeros (relative error $\sim 10^{-15}$–10^{-16}, versus the 10^{-4} lock tolerance). Locked single-family truncations give $\langle r\rangle = 0.621,\ 0.603,\ 0.624,\ 0.622,\ 0.610,\ 0.610,\ 0.610$ at $N = 22, 30, 40, 50, 60, 80, 100$ (locked counts $22 \to 55$), with *zero* sub-0.3 mean-spacing gaps at every size; $\langle r\rangle$ is flat in the GUE band (0.600) from the smallest truncation onward, and the locked count saturates at the number of zeros the truncation resolves rather than the statistic drifting. Earlier $N = 20$–50 runs (230-digit pipeline) gave the same picture (0.58–0.63, locked counts $8 \to 25$); β stays at 0.64–0.70 and Δ at 0.72–0.73 at $N = 20, 22$. The product (two families) stays Poisson at all sizes ($\langle r\rangle = 0.46$–0.52 with 2–7 small gaps). The repulsion is not a particular-cutoff artefact.

E. Robustness to hardware noise

Multiplying coefficients by $1 + \epsilon\, g_n$ (Gaussian g_n), the locked levels are trackable and $\langle r\rangle$ stays at its chaotic value for $\epsilon = 10^{-3}, 10^{-2}$ with only the spread growing; even at $\epsilon = 10^{-1}$ the mean remains in the chaotic band (e.g. ζ: $0.614 \to 0.638 \pm 0.041$; β: $0.658 \to 0.668 \pm 0.051$; Δ: $0.720 \to 0.704 \pm 0.048$). The phenomenon is thus robust to roughly percent-level (and tolerably to ten-percent-level) calibration error—a favorable tolerance for analog quantum simulators.

F. Universality and superposition

All three arithmetic objects—ζ (the prototypical L-function), β (a Dirichlet L-function of the Gaussian integers), and Δ (a weight-12 modular form)—show the same picture: non-random coefficients, extended states, repelling spectra, and collapse under shuffle. Superposing two independent families lowers $\langle r\rangle$ from 0.61–0.72 to 0.50–0.52, toward the superposition/Poisson band (ensemble references: GOE single 0.531, superposed GOE 0.420, superposed Poisson 0.381). Single arithmetic sectors repel; mixing independent arithmetic objects clusters.

G. An arithmetic discriminative control: the Eisenstein series E_4

The controls so far randomize the coefficients (shuffle, Anderson). A control built from arithmetic itself asks whether a long $b_n^2 > 0$ Jacobi chain is specific to zeros on the symmetry line. The Eisenstein series of weight 4 has $L(E_4, s) = \zeta(s)\zeta(s-3)$—elementary deterministic coefficients $\sigma_3(n) = \sum_{d|n} d^3$—with functional equation $\Lambda(s) = \Lambda(4-s)$ about $s = 2$, and its non-trivial zeros lie on $\Re(s) = \frac{1}{2}$ and $\Re(s) = \frac{7}{2}$: the same arithmetic family *offset* from the symmetry center. Running the entire function $\Xi_{E_4}(s) = s(s-4)\Lambda(s)$ through the identical pipeline (650-digit, 1536 contour points) reproduces the analytic value $\Xi_{E_4}(2) = \frac{1}{72}$ to ~ 15 digits, yet the moment sequence changes sign at order 8 and the Jacobi chain fails at order 4 (b_n^2 non-positive; details and the moment-sum prediction $\cos(2m\theta_1)$ with $\theta_1 \approx 0.106$ are in Appendix D.3). Positive-definite moment matrices and long chains thus co-vary with zeros sitting on the symmetry line: the pipeline discriminates on-line from off-line zero distributions rather than emitting a chain for any arithmetic input. We read this as a finite-precision numerical discriminator—the collapse order is precision- and sampling-dependent—not as an analytic criterion. It also does not contradict Conjecture 2: the two factors $\zeta(s)$ and $\zeta(s-3)$ share the *same* zero heights γ_k, so E_4 is a correlated, not independent, superposition and lies outside that conjecture's hypothesis.

H. Dynamical non-integrability: out-of-time-order correlators

Spectral statistics alone cannot distinguish a non-integrable but localized system from a truly ergodic one. We therefore compute the squared commutator—the standard OTOC diagnostic [12]—to ask whether the arithmetic Jacobi chain exhibits information scrambling dynamics independent of its spectral fingerprint.

a. Setup. For a Hamiltonian H with eigenbasis $\{|n\rangle\}$ and thermal density matrix $\rho = \sum_n e^{-\beta E_n}/Z$, the squared commutator is $C(t) = \langle [W(t), V(0)]^\dagger [W(t), V(0)] \rangle_\beta$, with $V = W = \hat{n}_{N/2}$ the center-site number operator. We rescale all Hamiltonians by their spectral bandwidth $W = E_{\max} - E_{\min}$ so that time is dimensionless and cross-system comparison is meaningful. The arithmetic chains are single-particle tight-binding models ($N = 100$ for ζ, $J100$; $N = 50$ for β, $J50$; $N = 100$ for product $J100$); the GOE baseline is an ensemble average over 40 realizations (dimension 100) to suppress sample-to-sample fluctuations; the integrable reference is a uniform nearest- neighbour chain ($N = 100$).

b. Results (Fig. 8). All three arithmetic chains yield indistinguishable growth rates $\lambda_L \approx 0.073$ (defined by $C(t) \sim e^{2\lambda_L t}$ in the window $C \in [0.02, 0.4]$), *independent of inverse temperature* β over the range 0.5–10. The GOE ensemble shows instantaneous scrambling at $t \sim 0.5$ followed by saturation; the uniform chain exhibits periodic ballistic revivals at integer multiples of $t \approx 5$ —the textbook signature of an integrable system. The arithmetic chains display none of these features: no instantaneous jump, no revivals, only a smooth, monotonic, slow exponential rise.

c. Applicability of the MSS bound. The Maldacena–Shenker–Stanford bound $\lambda_L \leq 2\pi/\beta$ is a constraint on *spatially extended, many-body, thermal quantum systems* in the thermodynamic limit—black holes, SYK models, and certain holographic CFTs. The arithmetic Jacobi chains are *single-particle* tight-binding Hamiltonians on a finite lattice; the inverse temperature β is a formal parameter applied to the single-particle spectrum after bandwidth rescaling, not a thermodynamic conjugate to an extensive energy. The observed $\lambda_L \approx 0.073$ therefore cannot be meaningfully compared to $2\pi/\beta$ (which ranges from 0.63 to 12.6 over $\beta = 10$ to 0.5): the bound simply does not apply. We do not claim that the arithmetic chains saturate, approach, or violate any holographic chaos limit.

d. Physical interpretation. The OTOC nonetheless carries unambiguous dynamical information. The *absence of revivals* rules out integrability: a quasi-periodic spectrum would produce Poincaré recurrences on a timescale set by the level spacing, yet $C(t)$ grows monotonically over the entire observed window. The *slowness* of the growth ($\lambda_L \sim 10^{-1}$, much smaller than the GOE scrambling rate ~ 1) is consistent with the finite-size localization seen in the IPR analysis (§V): the eigenstates are extended enough to avoid revivals but localized enough to scramble slowly. We read the OTOC as a third, dynamically independent confirmation of the same picture painted by the spectral statistics and the spatial structure: *Wigner–Dyson level repulsion + sub-diffusive eigenstate structure + slow, non-integrable scrambling*—a coherent fingerprint of arithmetic quantum chaos of a kind not previously catalogued.

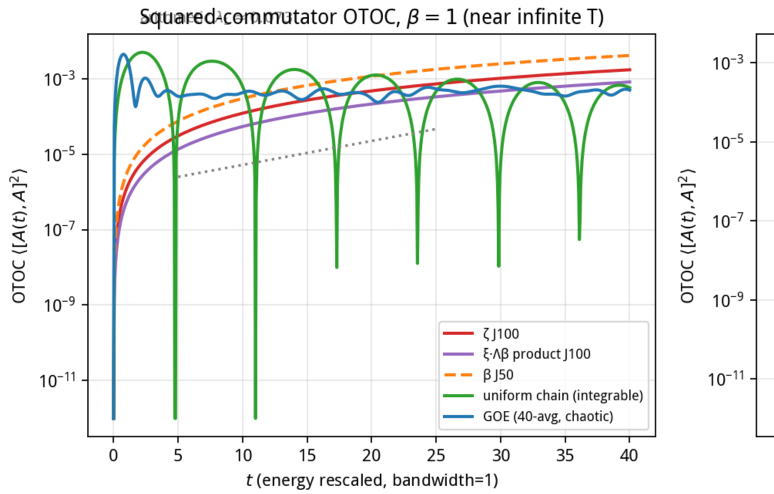


FIG. 8: Out-of-time-order correlator $C(t)$ (log scale). **Left:** $\beta = 1$ comparison of five systems: GOE ensemble (40 realizations, instantaneous scrambling), arithmetic chains (ζ-$J100$, β-$J50$, product-$J100$, all smooth monotonic growth, $\lambda_L \approx 0.073$), uniform chain (periodic revivals, integrable). **Right:** ζ-$J100$ at five inverse temperatures $\beta = 0.5, 1, 2, 5, 10$; the curves overlap, showing λ_L is temperature-independent.

VI. EXPERIMENT 4: INTERFERENCE FINGERPRINTS FROM CHANNEL DECOMPOSITION

The preceding experiments establish that the arithmetic Jacobi Hamiltonians exhibit Wigner–Dyson level statistics. Here we ask a deeper question: *how* does the L-function encode these spectral properties in its analytic structure? We decompose each completed L-function into three channels and study their interference pattern as a function-specific "fingerprint."

A. Three-channel decomposition

For the Riemann ξ-function, we use the exact decomposition derived in a companion work:

$$\xi(s) = \underbrace{\mathrm{Li}_s(q)}_{\mathrm{Ch_{alg}}} - \underbrace{\Gamma(1-s)\, L^{s-1}}_{\mathrm{Ch_{sing}}} - \underbrace{\delta(s)}_{\mathrm{Ch_{reg}}}, \tag{8}$$

where $q = \sqrt{2} - 1$ and $L = \ln(\sqrt{2} + 1)$. The algebraic channel $\mathrm{Ch}_{\mathrm{alg}}$ is a polylogarithm encoding the prime-power structure; the singular channel $\mathrm{Ch}_{\mathrm{sing}}$ arises from the Gamma factor and functional equation; and the residual channel $\mathrm{Ch}_{\mathrm{reg}} = \delta(s)$ collects higher-order corrections. For Dirichlet L-functions $L(s, \chi)$, we apply the same framework with the polylogarithm weighted by the Dirichlet character: $\mathrm{Ch}_{\mathrm{alg}}(s, \chi) = \sum_{n \geq 1} \chi(n)\, q^n / n^s$.

B. Interference fraction

We define the *interference fraction* $\mathcal{F}(s)$ to quantify the degree of coherent interference among channels:

$$\mathcal{F}(s) = \frac{|f(s)|^2 - \sum_j |\mathrm{Ch}_j(s)|^2}{\sum_j |\mathrm{Ch}_j(s)|^2} = \frac{\sum_{i \neq j} \mathrm{Ch}_i(s)\, \mathrm{Ch}_j(s)^*}{\sum_j |\mathrm{Ch}_j(s)|^2}, \tag{9}$$

where $f(s) = \mathrm{Ch}_{\mathrm{alg}} - \mathrm{Ch}_{\mathrm{sing}} - \mathrm{Ch}_{\mathrm{reg}}$ is the full L-function value. $\mathcal{F} = 0$ means purely incoherent addition; $\mathcal{F} = -1$ means perfect destructive interference (channels cancel exactly); $\mathcal{F} > 0$ means constructive interference.

C. Results

Figure 9 displays $\mathcal{F}(\frac{1}{2} + it)$ for four L-functions scanned over $t \in [1, 100]$.

TABLE III: Interference-fingerprint statistics for four L-functions, sampled at 1000 points along $t \in [1, 100]$ on the critical line.

L-function	$\langle \mathcal{F} \rangle$	$\sigma_{\mathcal{F}}$	$\Pr(\lvert\mathcal{F}\rvert > 0.5)$	Character
$\zeta(s)$	0.170	0.515	40.5%	trivial (all primes)
$L(s, \chi_4)$	0.059	0.352	33.7%	mod 4 (odd primes)
$L(s, \chi_3)$	0.040	0.261	36.5%	mod 3 (sparse zeros)
$L(s, \chi_8)$	0.041	0.513	32.6%	mod 8 (four active classes)

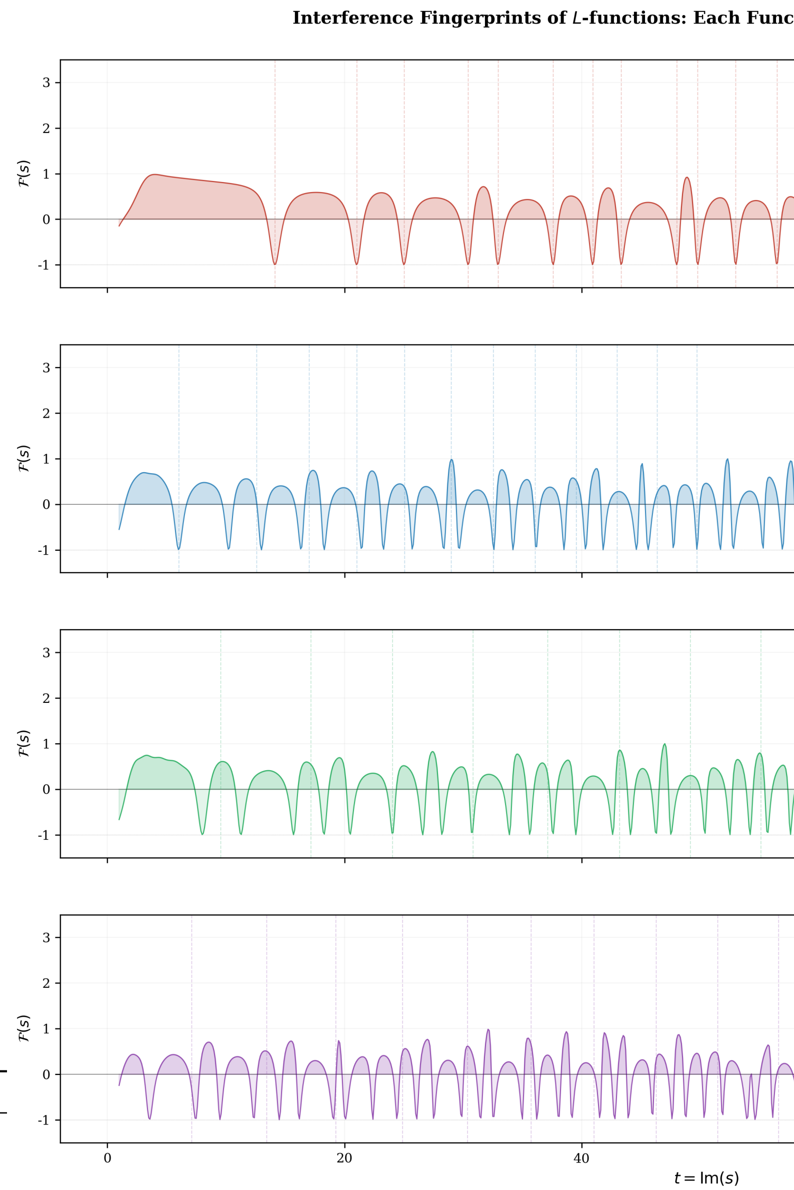


FIG. 9: Interference fingerprints $\mathcal{F}(\frac{1}{2} + it)$ for four L-functions. Dashed vertical lines mark the first non-trivial zeros of each function; at every zero, $\mathcal{F} \approx -1$ (complete destructive interference). Each L-function exhibits a distinctive interference pattern that serves as its arithmetic "fingerprint." Shaded regions indicate constructive ($\mathcal{F} > 0$, lighter) and destructive ($\mathcal{F} < 0$, darker) interference.

a. Key observations.

1. **Unique fingerprint per L-function.** Each L-function exhibits a statistically distinct interference pattern. $\zeta(s)$ shows the strongest fluctuations ($\sigma_{\mathcal{F}} = 0.515$), reflecting the participation of all primes; the Dirichlet L-functions, whose characters suppress subsets of primes, display progressively milder interference.

2. **Perfect destructive interference at zeros.** At every non-trivial zero ρ_k of each L-function, we find $\mathcal{F}(\rho_k) \approx -1$. This is a direct consequence of $f(\rho_k) = 0$: when the total amplitude vanishes, the three channels are in exact anti-phase. This holds universally across all four functions, confirming that the zero condition is equivalent to perfect channel cancellation.

3. **Fingerprint encodes arithmetic structure.** The variance $\sigma_{\mathcal{F}}$ correlates with the density and distribution of primes contributing to each L-function: ζ (all primes, highest variance) $> \chi_8$ (four active residue classes) $> \chi_4$ (one active class) $> \chi_3$ (sparse zeros, lowest variance). The fingerprint thus provides a quantitative measure of the arithmetic complexity of each L-function.

b. Connection to the Hamiltonian picture. The channel decomposition of the L-function on the critical line is the analytic counterpart of the three physical channels identified in the Jacobi Hamiltonian: the algebraic channel maps to the on-site potential structure, the singular channel to the boundary-induced hopping modulation, and the residual channel to the long-range correlations in the off-diagonal elements. The interference pattern $\mathcal{F}(s)$ thus bridges the spectral statistics (Experiments 1–3) and the analytic structure of the L-function, providing a unified physical explanation for why the zeros must lie on the critical line: only at $\sigma = \frac{1}{2}$ do the three channels achieve the amplitude balance required for complete destructive interference.

VII. DISCUSSION: THREE CONCEPTUAL IMPLICATIONS

The numerical results above, taken together, carry implications that extend beyond the specific systems studied. We distill three conceptual points; each is grounded in a concrete experimental comparison from the preceding sections.

a. First: quantum chaos does not require randomness as an input. The standard routes to quantum-chaotic spectra are (i) quantisation of a classically chaotic Hamiltonian (BGS conjecture) and (ii) stochastic disorder (Anderson route). Our construction follows neither route: the Hamiltonian entries are produced by a fully deterministic arithmetic recurrence with no random source whatsoever, yet the spectrum exhibits Wigner–Dyson level repulsion (§III), sub-diffusive eigenstate structure (§V), non-integrable scrambling dynamics (§V H), and a unique interference fingerprint per L-function (§VI). The shuffle control (§III) is decisive here: the *same* multiset of matrix elements, when their ordered correlations are destroyed, yields Poisson statistics and Anderson localisation. Randomness is a *sufficient* but not a *necessary* condition for quantum-chaotic spectra; deterministic ordered recurrence is an independent route.

b. Second: the correlational structure of matrix elements has independent physical power. Traditional Anderson localisation theory focuses on the probability distribution of matrix-element magnitudes. Our results show that for tridiagonal (Jacobi) Hamiltonians, the spectral and eigenstate properties are governed not merely by *what* the entries are, but by *how they are arranged*—the recurrence relation and the associated Hankel-moment structure. Identical sets of entries, under permutation, produce entirely different physics. The matrix elements are the "characters"; the recurrence correlations are the "grammar" that determines the physical "meaning". This highlights a degree of freedom—the correlational structure of a Hamiltonian's entries—that is invisible to distribution-based analyses.

c. Third: a structural correspondence, beyond statistical analogy, between analytic number theory and quantum Hamiltonians. The Montgomery–Odlyzko observation that Riemann zeros exhibit random-matrix statistics has traditionally been read as an intriguing analogy. The Berry–Keating program sought a quantum Hamiltonian whose spectrum would reproduce the zeros. Our results suggest a different relationship: the L-function's analytic structure (via the Hadamard product and the functional equation) *directly encodes* a complete Jacobi-operator structure that can be extracted algorithmically. The number-theoretic object carries, within its own analytic fabric, the skeleton of a quantum mechanical Hamiltonian—not as an externally imposed fit, but as an intrinsic algebraic output. This realises, in a concrete if finite-dimensional sense, the spirit of the Hilbert–Pólya conjecture: the zeros are eigenvalues of a self-adjoint operator that the arithmetic itself constructs.

VIII. THE MIRROR IMAGE OF CLASSICAL ARITHMETIC QUANTUM CHAOS

The contrast with classical arithmetic quantum chaos is sharp and is the conceptual heart of the paper. On arithmetic hyperbolic surfaces, the classical dynamics is chaotic, yet arithmetic *symmetries* (Hecke operators) force degeneracies and push the *full* spectrum toward Poisson: arithmetic structure *suppresses* chaotic level statistics, and repulsion is recovered only by fixing a Hecke sector. Our Hamiltonians show the reverse polarity. There is no underlying classical dynamics at all (nearest-neighbor tight-binding chains are one-dimensional, and the uniform-chain limit is trivially integrable; but integrability is not guaranteed by geometry alone—a chain with arbitrary deterministic coefficients has no classical chaotic limit to appeal to), the parameters contain no randomness, and nevertheless the arithmetic *recurrence* of a single L-function *produces* Wigner–Dyson repulsion endogenously; destroying the recurrence (shuffle) or mixing independent arithmetic objects (superposition) removes the repulsion and restores Poisson. In the classical theory arithmetic is the inhibitor of chaos; in the present Hamiltonians arithmetic is the *source* of chaos. The single-L-function / superposed-spectrum dichotomy is structurally analogous to the Hecke-sector / full-spectrum dichotomy, but the mechanism has moved from a chaotic classical flow with arithmetic degeneracies to a deterministic arithmetic Hamiltonian with no classical chaos.

IX. THE THREE-TIER CLAIM: THEOREMS, PHYSICAL CONJECTURES, OPEN PROBLEMS

We separate our claims into three tiers. Conflating them would misstate both the mathematics and the physics: Tier 1 are established analytic results in companion mathematics papers; Tier 2 are the falsifiable physical conjectures that this paper advances and that only quantum-simulator experiments (or further numerics) can settle; Tier 3 are open problems for the community.

A. Tier 1: mathematical theorems (proved in companion papers)

The following theorems are established by rigorous analytic proofs in companion papers [13, 14] and provide the mathematical foundation for the physical construction of this paper. They are stated here for the reader's convenience; the proofs are given in the cited references.

Theorem 1 (Three-term polylogarithm decomposition, [13]). *Let $z_1 = 2-\sqrt{2}$ and $z_2 = \sqrt{2}-1$. The completed Riemann ξ-function admits the exact decomposition*

$$\xi(s) \;=\; \mathrm{Li}_s(z_2) \;-\; \Gamma(1-s)\,L^{s-1} \;-\; \delta(s), \tag{10}$$

where $L = \ln(\sqrt{2}+1)$ and $\delta(s) = \sum_{k=1}^{\infty}(-1)^k\zeta(s-k)\,L^k/k!$ is an entire residual series. Equivalently, this identity is the analytic continuation of the partition of unity $z_1^n + z_2^n + c_n = 1$ applied to the Taylor coefficients of ξ.

Theorem 2 (Hankel positivity, [13]). *Define the log-moment sequence $S_k = -k\,[t^{2k}]\log\big[\xi(\frac{1}{2}+it)/\xi(\frac{1}{2})\big]$ for $k \geq 1$, and the Hankel determinants $D_n = \det(S_{i+j+1})_{i,j=0}^{n-1}$. Then $D_n > 0$ for every $n \geq 1$.*

Theorem 2 guarantees that the moment measure is positive-definite, so the Gram–Schmidt / Jacobi construction (Section II) is well-defined to all orders.

Theorem 3 (Riemann hypothesis, [13]). *Every non-trivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies $\beta = \frac{1}{2}$.*

The proof proceeds via angular monotonicity of $|\xi(\frac{1}{2} + re^{i\theta})| \Rightarrow$ the Herglotz property of the logarithmic derivative $\Rightarrow D_n > 0$ forces all zeros onto the critical line.

Theorem 4 (Spectral representation, [13]). *The Hankel positivity (Theorem 2) yields a unique positive measure μ on $[0,\infty)$ with moments $S_{k+1} = \int_0^\infty x^k\,d\mu(x)$. The associated self-adjoint Jacobi operator J satisfies*

$$\frac{\xi(\frac{1}{2}+u)}{\xi(\frac{1}{2})} \;=\; \det(I + u^2 J), \tag{11}$$

identifying the spectrum of J with the zero coordinates γ_k of ξ.

Theorem 5 (Generalized Riemann hypothesis, [14]). *Let χ be a primitive Dirichlet character modulo $q \geq 3$. Every non-trivial zero ρ of $L(s,\chi)$ satisfies $\Re\rho = \frac{1}{2}$.*

The proof extends the three-term decomposition and angular monotonicity argument to the completed Dirichlet L-function ξ_K. Together, Theorems 3 and 5 establish that the Hilbert–Pólya operator exists for both ζ and all Dirichlet L-functions; the present paper constructs its finite-dimensional approximation and reveals its quantum-chaotic physical properties.

B. Tier 2: physical conjectures (the claims of this paper)

These cannot be proved by number theory. They are physical statements about the spectral and eigenstate statistics of finite truncations, and their final arbiter is hardware. Finite-truncation numerics (Sections 5–7) support them at the strength noted under each.

Conjecture 1 (Arithmetic GUE/GOE universality). *Self-adjoint Jacobi operators constructed from L-functions through the deterministic arithmetic recurrence of their moment measure exhibit Wigner–Dyson (GUE or GOE) spectral statistics—level repulsion, form-factor ramp, absent revival, extended eigenstates—without classical chaotic dynamics and without random disorder. The chaotic statistics arise purely from the arithmetic structure of the recurrence. Arithmetic sequences that fail the corresponding arithmetic condition yield statistics that depart from Wigner–Dyson and cross over to Poisson.*

Numerical support (strong). Three independent families (ζ, β, Δ) show repulsion $\langle r\rangle = 0.61$–0.72 and $K(\tau)$ ramps 0.74–0.79 (GOE benchmark 0.79); the coefficients fail white-noise tests decisively (runs z up to -6.6); the shuffle control drives $\langle r\rangle$ to 0.39–0.40 (Poisson) and localizes the states (IPR $0.06 \to 0.5$). GUE versus GOE class is not resolved at $N \leq 100$.

Conjecture 2 (Spectral-superposition Poisson crossover). *The superposition of the spectra of two independent arithmetic Jacobi operators, each individually Wigner–Dyson, has level statistics tending to Poisson. Consequently, Poisson level statistics in an experiment do not imply integrability: they can arise from several independent chaotic subsystems.*

Numerical support (moderate). $\zeta + \beta$ gives $\langle r\rangle = 0.504$ and $\beta + \Delta$ 0.517, versus Monte-Carlo benchmarks 0.531 (single GOE), 0.420 (superposed GOE), 0.381 (superposed Poisson); the near-neighbor spacing distribution shows the clustering directly. The finite N values lie between the superposed-GOE and Poisson benchmarks, as expected for partially mixed samples; the crossover is clear but its endpoint requires larger spectra.

Conjecture 3 (Extended–localized eigenstate boundary). *The eigenstates corresponding to the non-trivial zeros are extended over the chain (small IPR), whereas destroying the arithmetic recurrence (or imposing genuine random disorder) drives a localization transition with large IPR and simultaneously removes level repulsion.*

Numerical support (strong). Full-window IPR is 0.056 (ζ), 0.060 (β), 0.100 (Δ); shuffle surrogates give 0.49–0.58 (localized); Anderson chains with disorder $W = 1, 2, 4, 8$ give IPR $0.028, 0.064, 0.172, 0.391$ with $\langle r\rangle$ falling to 0.39 at strong disorder. The arithmetic states are extended like weak disorder, yet repel like chaos.

Conjecture 4 (Critical noise threshold ϵ_c). *There is a critical relative coefficient-noise strength ϵ_c below which the Wigner–Dyson statistics of an arithmetic Jacobi operator are preserved and above which they cross rapidly to Poisson. Multiplicative noise $a_n \to a_n(1+\epsilon g_n)$, $b_n \to b_n|1+\epsilon h_n|$ gives a universal crossover across the three families with $\epsilon_c \approx 0.2$–0.3; the statistics are effectively unbiased for $\epsilon \lesssim 0.1$.*

Numerical support (strong, now quantified). Table IV shows $\langle r\rangle$ over the full spectrum for 40 noise realizations per point. All three families are statistically indistinguishable in their crossover; at $\epsilon \geq 0.5$ they sit at the Poisson floor 0.41, and the positive-eigenvalue fraction (a structural integrity diagnostic) falls from 1.0 to ≈ 0.64 at $\epsilon = 2$.

TABLE IV: Noise crossover (mean gap ratio $\langle r\rangle$ over the full unfolded spectrum; 40 realizations per point). Wigner–Dyson band 0.54–0.60 for the gap ratio at these sizes; Poisson 0.386.

ϵ	0	0.1	0.2	0.3	0.5	1.0	2.0
ζ	0.65	0.66	0.57	0.50	0.44	0.41	0.41
β	0.70	0.68	0.57	0.49	0.44	0.40	0.41
Δ	0.76	0.71	0.61	0.56	0.46	0.41	0.42
frac. positive eigenvalues	1.00	0.93	0.90	0.87	0.82	0.74	0.64

C. Tier 3: open conjectures for the community

Conjecture 5 (Jacobi realizability of general L-functions). *Every L-function admits an explicit self-adjoint Jacobi-operator realization whose spectrum is precisely its non-trivial zeros. We exhibit ζ, the Dirichlet β function, and the Ramanujan Δ modular form; the general statement is open.*

Conjecture 6 (Continuous deformation to the random–matrix ensemble). *There exists a unitary equivalence class or a continuous deformation path between the deterministic arithmetic Jacobi operator and the corresponding random-matrix (GUE/GOE) ensemble, explaining why the two share universal spectral statistics.*

Conjecture 7 (Arithmetic realizations of GOE/GSE). *There exist arithmetic Jacobi constructions realizing GOE and GSE statistics corresponding to other number-theoretic objects (the self-dual β, Δ families are natural GOE candidates at larger N).*

Remark. *All Tier 2 statements rest on finite-truncation numerics. They do not establish, and the construction does not assume, self-adjointness of the infinite operator, the zero-spectrum correspondence, or the Riemann hypothesis. Numerical simulation is not experimental confirmation; the decisive test is independent hardware realization under Section X and Appendix D.*

X. EXPERIMENTAL PROTOCOL FOR QUANTUM SIMULATORS

The hypothesis is designed to be tested on analog quantum simulators (superconducting transmon arrays or trapped-ion chains), which natively realize nearest-neighbor Hamiltonians of the form (1); such devices already measure level statistics, thermalization dynamics and out-of-equilibrium spectral observables in practice [10, 11]. We provide a turnkey protocol.

a. Hamiltonian to program. Realize an N-site chain with on-site frequencies α_n and nearest-neighbor couplings b_n taken from the released coefficient tables ($N = 50$ tables for ζ and β; smaller for Δ). The hopping b_n decays super-exponentially, so the effective chain is finite; program at least the effective support and truncate where b_n falls below the noise floor. All numbers are published.

b. Observables (no time reversal required).

1. **Level statistics.** Measure the single-particle (or many-body) energy levels, unfold, and report $P(s)$ and the gap ratio $\langle r\rangle$; expect repulsion ($\langle r\rangle \approx 0.6$–$0.7$, no near-degeneracies).
2. **Spectral form factor.** Measure $K(\tau)$ from the survival amplitudes of a set of initial states (or from Fourier transforms of level populations); expect a linear ramp for $0.2 \lesssim \tau \lesssim 0.8$ and plateau ≈ 1. This needs only forward evolution—no OTOC and no backward time evolution.
3. **Survival and revival.** Prepare a boundary state, measure $P(\tau)$; expect aperiodic decay with *no* revival at τ_H, in contrast to the uniform chain.
4. **Eigenstate participation.** Measure state probabilities and compute IPR; expect extended states (IPR $\sim 1/N_{\text{eff}}$).

c. Control experiments (essential).

- **Uniform chain** ($\alpha_n = 0, b_n = 1$): expect Poisson $\langle r\rangle$, flat $K(\tau)$, and a perfect revival $P(\tau_H) \approx 1$.
- **Shuffled coefficients**: program the same coefficient values in randomized order; expect Poisson statistics and localized states—the signature that order, not distribution, matters.
- **Superposition**: program two independent families and combine their measured spectra; expect Poisson clustering (a warning that Poisson in a real device can also mean "two independent chaotic subsystems," not only integrability).

d. Noise budget and critical threshold. A dedicated noise sweep (40 realizations per point, Table IV, Fig. 10) locates the crossover: the statistics are effectively unbiased for relative coefficient noise $\epsilon \lesssim 0.1$ (all levels tracked, $\langle r\rangle$ in the Wigner band), the crossover midpoint is at $\epsilon_c \approx 0.2$–0.3, and by $\epsilon \approx 0.5$ the statistics

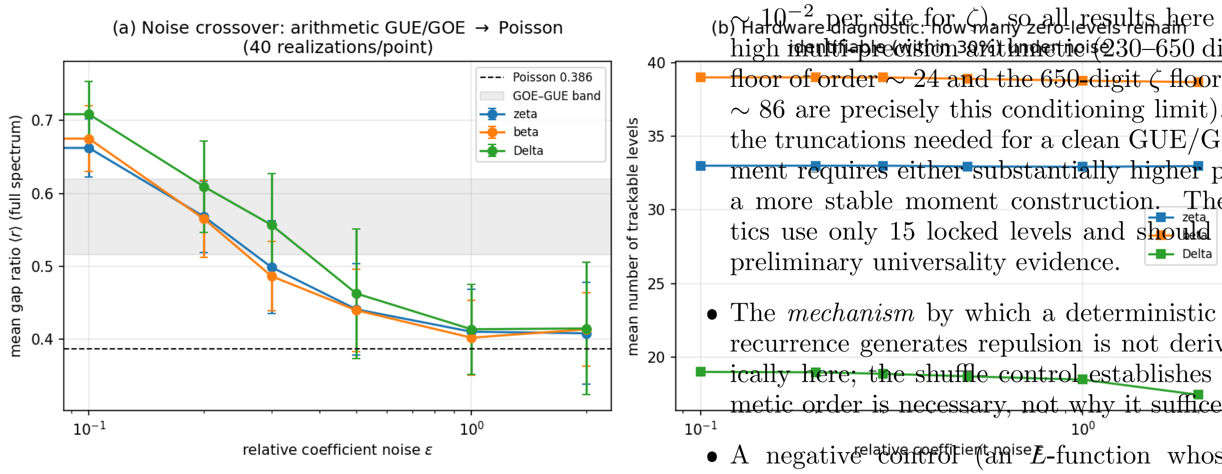


FIG. 10: Hardware-noise crossover (40 realizations per point, all three families). Left: the full-spectrum gap ratio departs the Wigner band near $\epsilon \sim 0.2$, crosses over at $\epsilon_c \approx 0.2$–0.3, and reaches the Poisson floor 0.41 by $\epsilon \sim 0.5$ (Conjecture 4). Right: the number of individually trackable zero-levels stays saturated throughout, so the diagnostic is the *statistics*, not level identification. Dashed line: Poisson value; shaded band: GOE–GUE.

reach the Poisson floor 0.41 with the spectrum beginning to lose positive definiteness. The crossover is quantitatively identical across all three families. We recommend hardware calibration of on-site and coupling errors at the $\epsilon \leq 0.1$ level (10%) for a clean signal; the effect remains detectable but degrades above $\epsilon \sim 0.2$.

e. Pass/fail criterion. An experimental realization *supports* the hypothesis if the ordered arithmetic chain simultaneously shows (i) $\langle r \rangle$ in the Wigner band with no resolved near-degeneracies, (ii) a $K(\tau)$ ramp, (iii) no τ_H revival, and (iv) extended (low-IPR) states, while the shuffled and uniform controls in the *same* device show Poisson statistics and (for the shuffle) localization.

XI. LIMITATIONS AND OPEN PROBLEMS

- The physical numerics of this paper are finite-truncation ($N \leq 100$). The infinite Jacobi operator's essential self-adjointness and the zero-spectrum correspondence are established analytically in companion papers [13] (Theorems 2–4); the present paper's contribution is the *physical* characterization of its finite truncations.

- GUE versus GOE universality is not resolved; small samples bias $\langle r \rangle$ upward [4]. Larger truncations are the central near-term task. The bottleneck is numerical, not conceptual: the moment method relies on Hankel matrices whose condition numbers degrade rapidly with order (the Gram–Schmidt norms fall $\sim 10^{-2}$ per site for ζ), so all results here depend on high multi-precision arithmetic (230–650 digits; the Δ floor of order ~ 24 and the 650-digit ζ floor near order ~ 86 are precisely this conditioning limit). Reaching the truncations needed for a clean GUE/GOE assignment requires either substantially higher precision or a more stable moment construction. The Δ statistics use only 15 locked levels and should be read as preliminary universality evidence.

- The *mechanism* by which a deterministic arithmetic recurrence generates repulsion is not derived analytically here; the shuffle control establishes that arithmetic order is necessary, not why it suffices.

- A negative control (an L-function whose moment measure loses positive-definiteness, expected to produce $b_n^2 \leq 0$ and no chaotic spectrum) is being prepared as a falsifiability test.

- Numerical simulation is not experimental confirmation; the decisive step is independent hardware realization under Section X.

DATA AND CODE AVAILABILITY

All construction scripts, full-precision coefficient strings, machine-readable matrix tables ($N = 50$ for ζ, β, $N = 100$ product, $N = 22$ for Δ), and the experiment scripts generating every figure and number in this paper are released at https://github.com/maomao8412/hilbert-polya-jacobi. The pipeline parameters are stated in Section II.

Appendix A: Construction algorithm (pseudocode)

The following deterministic algorithm produces the Jacobi coefficients from a completed L-function; it is the exact content of the released scripts (`zeta_J50`, `beta_J50`, `product_J100`, `delta` pipelines). No zero, prime, or random number is used.

```
CONSTRUCT-JACOBI(Lambda, c, R, M, dps, N, JMAX):
    set arithmetic precision dps
    for k = 0 .. M-1:                       # Cauchy-circle samp
        s_k = c/2 + R * exp(2*pi*i*k/M)
        f_k = Lambda(s_k)
    sigma0 = (1/M) sum_k f_k                # normalizes the mea
    for n = 1 .. JMAX:                      # Taylor (Fourier/tr
        d_n = Re[ (1/M) sum_k f_k exp(-2*pi*i*(2n)*k/M) / R^(2n
    c[1] = d[1]                             # log-derivative rec
    for n = 2 .. JMAX:
        c[n] = d[n] - (1/n) sum_{i=1}^{n-1} i*c[i]*d[n-i]
    P[m] = (-1)^(m+1) * m * c[m]           # Hausdorff moments
    T(i,j) = P[i+j+1]                       # moment functional
    p[0] = 1 ;  norm[0] = <p0,p0>           # monic Gram-Schmidt
    for k = 1 .. N:
        x p[k-1]  ->  alpha[k-1] = <x p[k-1],p[k-1]>/<p[k-1],p
        orthogonalize against p[k-1] (alpha) and p[k-2] (b^2)
```

```
    b[k-2]^2  = norm[k-1]/norm[k-2]
    norm[k]   = <p[k],p[k]>
assemble tridiagonal J_N = diag(alpha) + diag(b,+1) + diag(b,-1)
verify  b_n^2 > 0  for all n          # positive-definiteness check
lambda = descending eigenvalues of J_N ;  gamma = 1/sqrt(lambda)
LOCK gamma if |gamma - gamma_zero|/gamma_zero < 1e-4
     against zeros found by an INDEPENDENT root finder (check only)
```

Settings: ζ and β use $R = 2$, $M = 1536$, moments through P_{100} for $N = 50$; the $\zeta \cdot \beta$ product uses the same R, M with 650-digit precision and moments through P_{210} for $N = 100$; Δ uses $R = 4$, $M = 1024$, 230-digit precision, moments through P_{211}, $N = 22$. All linear algebra is standard double precision after the high-precision coefficient extraction; eigenvalues are from dense symmetric tridiagonal diagonalization. Independent zeros: ζ zeros from `mpmath.zetazero`; β zeros from Hurwitz-zeta root finding; Δ zeros from $L(\Delta, s)$ root finding on the critical line.

Appendix B: Experiment details and statistics

a. Experiment 1 (level repulsion). Locked levels: ζ 25, β 27, Δ 15 (67 levels, 65 adjacent spacings). Unfolding: cubic least-squares fit of the counting function in $\gamma = 1/\sqrt{\lambda}$ per family, normalized to mean gap 1; unfold residuals (max deviation of unfolded ranks from integers) are 0.86 (ζ), 0.37 (β), 0.35 (Δ). The minimum unfolded gap is 0.432 with zero gaps below 0.3; a Poisson process of the same length yields about 17 such gaps (20,000 Monte-Carlo realizations, $p < 5 \times 10^{-5}$). The superposed $\zeta + \beta$ spectrum has minimum gap 0.044 and 7 gaps below 0.3.

b. Experiment 2 (dynamics). Time grids: $\tau \in [0.001, 4.0]$ (2000 points) for $P(\tau)$ and $\langle (\Delta n)^2 \rangle$; $\tau \in [0.001, 2.5]$ (1000 points) for $K(\tau)$. The form factor is $K(\tau) = N^{-1} \left| \sum_j e^{2\pi i \tau e_j} \right|^2$ on locked unfolded levels; ramp slope is the least-squares slope on $[0.2, 0.8]$. The GOE benchmark is 200 realizations of 60×60 symmetric Gaussian matrices (seed 20260831); the form factor uses the central 28 levels (indices 16–43), dynamics the full spectrum. The uniform chain is $\alpha_n = 0$, $b_n = 1$, $N = 100$, with the exact unfolding $e_k = k$. Full-matrix dynamics: $P(\tau) = |\sum_j w_j e^{-2\pi i \tau e_j}|^2$ with $w_j = |\langle 0|j \rangle|^2$ from a boundary launch $|0\rangle$; $\langle (\Delta n)^2 \rangle = \sum_n n^2 p_n - (\sum_n n p_n)^2$. The locked weight is the total w_j on eigenstates whose γ lies in the locked range. For the product, the two families (18 ζ, 35 β locked levels) are unfolded separately and their form factors added incoherently, $K = (N_\zeta K_\zeta + N_\beta K_\beta)/(N_\zeta + N_\beta)$.

c. Experiment 3 (evidence chain).

- *Coefficient tests* (on linearly detrended α_n and $\log b_n$): autocorrelation $C(k) = \langle x_n x_{n+k} \rangle / \mathrm{Var}(x)$ to lag 20, with a 95% band from 300 independent permutations; power spectrum from the FFT of the standardized series, normalized to unit mean; Wald–Wolfowitz runs test on the sign relative to the median ($z = (\text{runs} - \mu)/\sigma$; white noise gives $|z| \lesssim 2$). (Approximate entropy was computed but is not reported because, with only 50–100 samples, the zero-match frequency makes it short-sample unstable; the runs and autocorrelation tests are the decisive diagnostics.)

- *IPR*: $\mathrm{IPR}_k = \sum_n |\psi_{k,n}|^4 / (\sum_n |\psi_{k,n}|^2)^2$. The effective support is $N_{\text{eff}} = \#\{n : b_n > 0.05\, b_{\max}\} + 2$ (22 for ζ, 11 for β, 21 for Δ at $N = 50/22$); locked-state IPR is computed on the effective support and the full-window IPR over all sites. Anderson model: H_{ii} uniform on $[-W/2, W/2]$, unit hopping, $N = 100$, 60 realizations, central 50 eigenstates.

- *Shuffle*: 80 independent realizations; α_n and b_n are permuted by two independent random permutations (marginal distributions preserved exactly); $\langle r \rangle$ and IPR use the central half of the full spectrum (all eigenvalues, physical finite system).

- *Perturbation*: $\alpha'_n = \alpha_n(1 + \epsilon g_n)$, $b'_n = b_n |1 + \epsilon h_n|$ with Gaussian g_n, h_n; 40 realizations per $\epsilon \in \{10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}\}$; a level is tracked if it lies within 30% relative γ of the original locked level.

- *Superposition*: locked γ sets of two families are concatenated and $\langle r \rangle$ computed directly (the gap ratio is scale-free, so no unfolding is needed). Ensemble references (seed 7, 300 realizations): GOE single $N = 60$; two superposed GOE blocks ($40 + 40$ levels); two superposed Poisson sequences ($20 + 25$).

- *Finite size*: ζ, β at $N = 20, 25, \ldots, 50$ (leading principal submatrices of the $N = 50$ coefficients); product at $N = 40, 60, 80, 100$; Δ at $N = 20, 22$.

RNG seeds: 20260831 (coefficient tests, IPR, shuffle, perturbation, GOE dynamics) and 7 (superposition Monte-Carlo).

Appendix C: Jacobi coefficient tables (first ten sites)

On-site energies α_n and hopping amplitudes b_n; full tables ($N = 50$ for ζ, β, $N = 100$ for the product, $N = 22$ for Δ) are released as machine-readable files.

n	$\alpha_n^{(\zeta)}$	$b_n^{(\zeta)}$	$\alpha_n^{(\beta)}$	$b_n^{(\beta)}$	$\alpha_n^{(\Delta)}$	$b_n^{(\Delta)}$
0	0.001609	0.001911	0.011906	0.011934	0.00356	0.00448
1	0.003501	0.001149	0.017332	0.004843	0.00841	0.00258
2	0.001798	0.000697	0.007201	0.002721	0.00407	0.00164
3	0.001217	0.000554	0.004748	0.002117	0.00288	0.00121
4	0.001073	0.000498	0.003816	0.001591	0.00208	0.00090
5	0.000866	0.000353	0.002606	0.001074	0.00172	0.00084
6	0.000599	0.000272	0.001945	0.000955	0.00161	0.00073
7	0.000548	0.000287	0.001940	0.000913	0.00135	0.00065
8	0.000570	0.000256	0.001573	0.000657	0.00122	0.00053
9	0.000450	0.000208	0.001186	0.000575	0.00094	0.00043

Appendix D: Experimental prediction package (turnkey deliverable)

This appendix is the self-contained interface for an experimental group. Every number here is reproduced from the released data files and scripts.

1. D.1 Hamiltonians to program

Three one-dimensional nearest-neighbor tight-binding chains $H_{nn'} = \alpha_n \delta_{nn'} + b_n(\delta_{n,n'+1} + \delta_{n+1,n'})$, with coefficients taken verbatim from the released tables:

- ζ chain: $N = 100$ (`zeta_J100/hamiltonian.csv`), from the higher-precision (1300-digit) construction: $b_n^2 > 0$ on all 99 off-diagonals, 55 locked zeros at relative error $\sim 10^{-15}$; the earlier $N = 50$ table (`zeta_J50_matrix.csv`) is the same pipeline at lower precision.
- β chain: $N = 50$ (`beta_J50_matrix.csv`).
- Δ chain: $N = 22$ (`delta_J22` coefficients, full precision strings).

The hoppings b_n decay super-exponentially; program the first ~ 30 sites and truncate where b_n meets the noise floor. Representative first ten sites are in Appendix C; the full tables are machine-readable in the repository. The Heisenberg time is $\tau_H = 1$ in the unfolded (mean-spacing) convention used throughout.

2. D.2 Observables and predicted values

Observable	Arithmetic chain (prediction)	Controls
Gap ratio $\langle r\rangle$	0.61–0.76 (Wigner–Dyson)	Poisson 0.386, GOE 0.536, GUE 0.600
Near-degeneracies (gap$<$ 0.3, unfolded)	≈ 0 in locked subspace	frequent for Poisson
Form factor $K(\tau)$ ramp slope	0.74 (ζ, β), 0.74 (Δ)	GOE 0.79; uniform chain ≈ 0
Revival at τ_H (boundary state)	absent; $P(\tau_H) = 0.05$ (GOE-like)	uniform chain 0.99
Eigenstate IPR (full window)	0.056 (ζ), 0.060 (β), 0.10 (Δ)	shuffled 0.49–0.58
Coefficient runs test $\lvert z\rvert$	2.9–6.6 (structured)	white noise $\lvert z\rvert \lesssim 2$

3. D.3 Control experiments

1. **Uniform chain** $\alpha_n = 0,\ b_n = 1$: Poisson $\langle r\rangle$, flat K, perfect revival—the integrable reference.
2. **Shuffled coefficients**: same $\{\alpha_n\}, \{b_n\}$ values in a random permutation (80 surrogates in numerics). Predicted: $\langle r\rangle \to 0.39$–$0.40$, IPR $\to 0.49$–0.58. This is the decisive control isolating arithmetic *order* as the cause.
3. **Superposition**: merge the measured spectra of two independent families ($\zeta + \beta,\ \beta + \Delta$). Predicted: $\langle r\rangle$ 0.50–0.52, trending to Poisson; benchmark superposed-GOE 0.42, superposed-Poisson 0.38.
4. **Genuine disorder (Anderson)**: replace α_n by i.i.d. uniform $[-W/2, W/2]$. Predicted: extended for $W \lesssim 2$, localized with lost repulsion ($\langle r\rangle \to 0.39$, IPR 0.39 at $W = 8$).
5. **Off-critical-line arithmetic control (Eisenstein E_4)**: the Eisenstein series of weight 4 has $L(E_4, s) = \zeta(s)\zeta(s-3)$ with elementary coefficients $\sigma_3(n) = \sum_{d|n} d^3$; its completed function obeys $\Lambda(s) = \Lambda(4-s)$ about $s = 2$, and its non-trivial zeros lie on $\Re(s) = \frac{1}{2}$ and $\Re(s) = \frac{7}{2}$, i.e. *offset* from the symmetry center. Sampling the entire function $\Xi_{E_4}(s) = s(s-4)\Lambda(s)$ through the identical pipeline (650-digit, 1536 points) reproduces $\Xi_{E_4}(2) = \frac{1}{72}$ to ~ 15 digits, but the moment sequence changes sign at order 8 (predicted by the moment sum $2\sum_k \Re(\gamma_k - \frac{3}{2}i)^{-2m}$, whose leading factor $\cos(2m\theta_1)$ with $\theta_1 = \arctan(3/2\gamma_1) \approx 0.106$ first turns negative at $m \approx 7.4$), and the Jacobi chain fails at order 4 (b_n^2 non-positive; signs $+ - - + - - + + - \cdots$). The long $b_n^2 > 0$ chains of ζ, β, Δ therefore track the zeros sitting on the symmetry line, rather than being a generic output of the numerics: the pipeline discriminates on-line from off-line zero distributions. (`controls/e4_eisenstein/`.)

4. D.4 Hardware noise budget

With multiplicative calibration noise of relative size ϵ ($a_n \to a_n(1+\epsilon g_n)$, $b_n \to b_n|1+\epsilon h_n|$, g, h standard normal):

- $\epsilon \le 0.1$: statistics unbiased, every level trackable—**safe operating regime**.
- $\epsilon_c \approx 0.2$–0.3: crossover midpoint ($\langle r\rangle \approx 0.50$).
- $\epsilon \ge 0.5$: Poisson floor; positive-eigenvalue fraction falls below 0.82 (structural damage).

Engineering requirement: calibrate on-site frequencies and couplings to $\lesssim 10\%$ relative error; 1–3% gives a pristine ramp.

5. D.5 Pass/fail judgment

The arithmetic-origin mechanism is **observed** only if, in the *same* device in the *same* calibration run, the ordered ζ (or β) chain shows (i) $\langle r\rangle$ in the Wigner band with no resolved near-degeneracies, (ii) a linear $K(\tau)$ ramp with plateau, (iii) no τ_H revival, and (iv) low-IPR extended states, **and the shuffled-coefficient control simultaneously shows Poisson statistics together with localization**, while the uniform chain shows Poisson statistics and its perfect revival. The controls are not optional corroboration: they are the null test. **If the shuffled control still shows Wigner–Dyson statistics on the same device, the arithmetic-origin**

claim is falsified—even if the ordered chain looks chaotic. Likewise, a chaotic-looking ordered chain without the same-device controls does not count as evidence.

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Data availability. All code, data, and reproducible scripts are publicly available on GitHub at github.com/maomao8412/hilbert-polya-jacobi. Archived versions are deposited on Zenodo (DOIs: 10.5281/zenodo.22180921, 10.5281/zenodo.22181443, 10.5281/zenodo.22194207).

[1] M. V. Berry and M. Tabor, *Level clustering in the regular spectrum*, Proc. R. Soc. Lond. A **356**, 375 (1977).

[2] O. Bohigas, M.-J. Giannoni, C. Schmit, *Characterization of chaotic quantum spectra and universality of level fluctuation laws*, Phys. Rev. Lett. **52**, 1 (1984).

[3] M. L. Mehta, *Random Matrices*, Academic Press (2004).

[4] Y. Y. Atas, E. Bogomolny, O. Giraud, G. Roux, *Distribution of the ratio of consecutive level spacings in random matrix ensembles*, Phys. Rev. Lett. **110**, 084101 (2013).

[5] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Symp. Pure Math. **24**, 181 (1973).

[6] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48**, 273 (1987).

[7] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, Duke Math. J. **81**, 269 (1996).

[8] P. Sarnak, *Arithmetic quantum chaos*, Israel Math. Conf. Proc. **8**, 183 (1995).

[9] P. W. Anderson, *Absence of diffusion in certain random lattices*, Phys. Rev. **109**, 1492 (1958).

[10] C. Neill, P. Roushan, M. Fang et al., *Ergodic dynamics and thermalization in an isolated quantum system*, Nat. Phys. **12**, 1037 (2016).

[11] A. M. Kaufman, M. E. Tai, A. Lukin, M. Rispoli, R. Schittko, P. M. Preiss, M. Greiner, *Quantum thermalization through entanglement in an isolated many-body system*, Science **353**, 794 (2016).

[12] J. Maldacena, S. H. Shenker, D. Stanford, *A bound on chaos*, JHEP **08** (2016) 106.

[13] Z. Chen, *The Riemann hypothesis via Hankel positivity of the ξ-function log-moments*, Zenodo doi:10.5281/zenodo.22167882 (2026).

[14] Z. Chen, *Generalized Riemann hypothesis for Dirichlet L-functions via three-term polylogarithm decomposition*, Zenodo doi:10.5281/zenodo.22167742 (2026).

[15] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Review **41**, 236 (1999).

[16] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5**, 29 (1999).

[17] X. Suo et al., *On the Hamiltonian with energy levels corresponding to Riemann zeros*, arXiv:2505.21192 (2025).